EM-Normal mixture

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$$\theta = (\pi_1, \pi_2, \pi_{K-1}, \mu_1, \cdots \mu_K, \sigma_1^2, \cdots \sigma_K^2)$$

$$p(X_1, \cdots X_n, Z_1, \cdots Z_n \mid \theta)$$

$$= \prod_{i=1}^{n} p(X_i \mid Z_i, \theta)\, p(Z_i \mid \theta) = \prod_{i=1}^{n} \prod_{j=1}^{K} \left[p(X_i \mid Z_i=j, \theta)\, p(Z_i=j \mid \theta) \right]^{I(Z_i=j)}$$

$$\log p(X_1, \cdots X_n, Z_1, \cdots Z_n \mid \theta)$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{K} I(Z_i=j) \left[\log p(X_i \mid Z_i=j, \theta) + \log p(Z_i=j \mid \theta) \right]$$

$$= \sum_{i=1}^{n} \left[\sum_{j=1}^{K} I(Z_i=j) \left(-\log \sqrt{2\pi\sigma_j^2} - \frac{1}{2} \frac{(x_i-\mu_j)^2}{\sigma_j^2} + \log \pi_j \right) \right]$$

$$\pi_K = 1 - (\pi_1 + \pi_2 + \cdots \pi_{K-1})$$

$$Q(\theta, \theta^\circ) = E_{Z \mid X, \theta^\circ} \left[\log p(X_1, \cdots X_n, Z_1 \cdots Z_n \mid \theta) \right]$$

$$= \sum_{i=1}^{n} \left[\sum_{j=1}^{K} p(Z_i=j \mid X, \theta^\circ) \left(-\log \sqrt{2\pi\sigma_j^2} - \frac{1}{2} \frac{(x_i-\mu_j)^2}{\sigma_j^2} + \log \pi_j \right) \right]$$

E-step:

$$j = 1, 2, \cdots K \qquad \theta^\circ = (\pi_1^\circ, \pi_2^\circ, \pi_{K-1}^\circ, \mu_1^\circ, \cdots \mu_K^\circ, \sigma_1^{2\circ} \cdots \sigma_K^{2\circ})$$

$$p(Z_i=j \mid X, \theta^\circ) = \frac{p(Z_i=j, X_i \mid \theta^\circ)}{p(X_i \mid \theta^\circ)}$$

$$= \frac{p(X_i \mid Z_i=j, \theta^\circ)\, p(Z_i=j \mid \theta^\circ)}{\sum_{k=1}^{K} p(X_i \mid Z_i=k, \theta^\circ)\, p(Z_i=k \mid \theta^\circ)}$$

$$= \frac{\frac{1}{\sqrt{2\pi\sigma_j^{2\circ}}} \exp\left(-\frac{(x_i-\mu_j^\circ)^2}{2\sigma_j^{2\circ}} \right) \pi_j^\circ}{\sum_{k=1}^{K} \frac{1}{\sqrt{2\pi\sigma_k^{2\circ}}} \exp\left(-\frac{(x_i-\mu_k^\circ)^2}{2\sigma_k^{2\circ}} \right) \pi_k^\circ}$$

$$\pi_K^\circ = 1 - (\pi_1^\circ + \pi_2^\circ \cdots + \pi_{K-1}^\circ)$$

M-step: $j = 1, 2, \cdots K-1$

$$\frac{\partial Q(\theta, \theta^\circ)}{\partial \pi_j} = \sum_{i=1}^{n} \left[\frac{p(Z_i=j \mid X, \theta^\circ)}{\pi_j} - \frac{p(Z_i=K \mid X, \theta^\circ)}{1 - (\pi_1 + \pi_2 + \cdots \pi_{K-1})} \right]$$

$$= \frac{[1 - (\pi_1 + \pi_2 + \cdots \pi_{K-1})] \left[\sum_{i=1}^{n} p(Z_i=j \mid X, \theta^\circ) \right] - \pi_j \left[\sum_{i=1}^{n} p(Z_i=K \mid X, \theta^\circ) \right]}{\pi_j [1 - (\pi_1 + \pi_2 + \cdots \pi_{K-1})]} = 0$$

$$j = 1, 2, \cdots K-1$$

$$\underbrace{[1 - (\pi_1 + \pi_2 + \cdots \pi_{K-1})]}_{\pi_K} \left[\sum_{i=1}^{n} \underbrace{p(Z_i=j \mid X, \theta^\circ)}_{A_{ij}} \right] - \pi_j \left[\sum_{i=1}^{n} \underbrace{p(Z_i=K \mid X, \theta^\circ)}_{A_{iK}} \right] = 0$$

$$= 1 - \sum_{j=1}^{K-1} A_{ij}$$

$$j=1 \qquad \pi_K \sum_{i=1}^{n} A_{i1} = \pi_1 \sum_{i=1}^{n} A_{iK}$$

$$j=2 \qquad \pi_K \sum_{i=1}^{n} A_{i2} = \pi_2 \sum_{i=1}^{n} A_{iK}$$

$$j=K-1 \qquad \pi_K \sum_{i=1}^{n} A_{iK-1} = \pi_{K-1} \sum_{i=1}^{n} A_{iK}$$

sum

$$\pi_K \sum_{i=1}^{n} \sum_{j=1}^{K-1} A_{ij} = (1 - \pi_K) \sum_{i=1}^{n} A_{iK}$$

$$\pi_K \sum_{i=1}^{n} 1 = \sum_{i=1}^{n} A_{iK} \qquad \pi_K = \frac{\sum_{i=1}^{n} A_{iK}}{n}$$

$$\hat{\pi}_j = \frac{1}{\sum_{i=1}^{n} A_{iK}} \pi_K \sum_{i=1}^{n} A_{ij} = \frac{1}{\cancel{\sum_{i=1}^{n} A_{iK}}} \frac{\cancel{\sum_{i=1}^{n} A_{iK}}}{n} \cdot \sum_{i=1}^{n} A_{ij}$$

$$j = 1, \cdots, K$$

$$\boxed{\begin{aligned} Q(\theta, \theta^\circ) &= E_{Z \mid X, \theta^\circ} \left[\log p(X_1, \cdots X_n, Z_1 \cdots Z_n \mid \theta) \right] \\ &= \sum_{i=1}^{n} \left[\sum_{j=1}^{K} p(Z_i=j \mid X, \theta^\circ) \left(-\log \sqrt{2\pi\sigma_j^2} - \frac{1}{2} \frac{(x_i-\mu_j)^2}{\sigma_j^2} + \log \pi_j \right) \right] \end{aligned}}$$

$$\frac{\partial Q(\theta, \theta^\circ)}{\partial \mu_j} = \sum_{i=1}^{n} p(Z_i=j \mid X, \theta^\circ) \left[-\frac{\cancel{2}(x_i-\mu_j)}{\cancel{2}\sigma_j^2} \times (-1) \right]$$

$$= \frac{1}{\sigma_j^2} \left[\sum_{i=1}^{n} p(Z_i=j \mid X, \theta^\circ)\, x_i - \mu_j \sum_{i=1}^{n} p(Z_i=j \mid X, \theta^\circ) \right] = 0$$

$$\hat{\mu}_j = \frac{\sum_{i=1}^{n} p(Z_i=j \mid X, \theta^\circ)\, x_i}{\sum_{i=1}^{n} p(Z_i=j \mid X, \theta^\circ)}$$

$$j = 1, \cdots, K \qquad -\log(\sigma_j^2)^{\frac{1}{2}} = -\frac{1}{2} \log \sigma_j^2$$

$$\frac{\partial Q(\theta, \theta^\circ)}{\partial \sigma_j^2} = \sum_{i=1}^{n} p(Z_i=j \mid X, \theta^\circ) \left(-\frac{1}{2} \frac{1}{\sigma_j^2} - \frac{1}{2} (x_i-\mu_j)^2 \left(-\frac{1}{(\sigma_j^2)^2} \right) \right)$$

$$= \sum_{i=1}^{n} p(Z_i=j \mid X, \theta^\circ) - \frac{1}{2} \frac{1}{(\sigma_j^2)^2} \left(\sigma_j^2 - (x_i-\mu_j)^2 \right) = 0$$

$$\hat{\sigma}_j^2 = \frac{\sum_{i=1}^{n} p(Z_i=j \mid X, \theta^\circ)(x_i-\mu_j)^2}{\sum_{i=1}^{n} p(Z_i=j \mid X, \theta^\circ)}$$